



NASA's open source fine-print

NASA advised to restructure their open source software and permissive license requirements. <https://digit.in/nov21-45>



Amazon and internet domination

Via its new Project Kuiper, Amazon plans to launch its first internet satellites by late 2022. <https://digit.in/nov21-47>

Apple iPhone 13

A refined upgrade

Apple's iPhone 13 is a refined upgrade to last year's iPhone 12 bringing in all-around improvements with a new camera, faster chip, larger battery and a smaller notch. We took the iPhone 13 for a spin to see how its upgrades stack up.

BUILD AND DESIGN

Although the iPhone 13 is a bit thicker than the iPhone 12, it follows the same design principles. It has an aluminium frame with a Ceramic shield glass, making it tough and protects it against drops. The iPhone 13 is also IP68 dust and water protected.

DISPLAY

The 13 features a 6.1-inch XDR OLED display which is superb for media consumption. It is vibrant, gets quite bright and the Face ID notch is now smaller than before.

Using the Calman and SpectraCal C6 colourimeter revealed that, the display has a peak brightness of 536 nits. Further testing showed that it has an average delta of 1.9 in the sRGB colour space. In daily use, the iPhone 13 offers a superb viewing experience.

SPECS AND FEATURES

The iPhone 13 is powered by the A15 Bionic chip made on 5nm architecture with two high-performance cores and four power-efficient cores. Apple has used new camera sensors on the iPhone 13 headlined by a 12MP camera with f/1.6 aperture and sensor-shift OIS followed by a 12MP ultra-wide camera.

PERFORMANCE

The iPhone 13 has the A15 Bionic chip with a hexa-core CPU, quad-core GPU and 16-core neural engine for machine learning, all powering the iOS 15 experience. The iPhone 13 demonstrates superiority in synthetic benchmarks over its Android



Price
₹79,900

rivals. In AnTuTu tests, the iPhone 13 received high scores leaving behind phones like the OnePlus 9 Pro. It also topped the charts in Geekbench CPU tests.

The iPhone 13 aced GPU tests like 3DMark Wildlife and GFXBench Metal that prove its mettle in running graphic-intensive games. We used Gamebench to record gameplay metrics where the iPhone 13 runs COD Mobile at 59 FPS at 100% stability.

All of this just goes on to highlight the great real world performance that you can expect from the iPhone 13.

CAMERA

Instead of adding high-resolution sensors, the iPhone 13 instead relies on ML algorithms and optimizations to produce natural-looking pictures. But, because a phone like the OnePlus 9 Pro uses a high-resolution sensor, you get much better detail and sharpness which is realised as you zoom in.

One of the highlights of the cameras this year is the Cinematic Mode which brings depth-of-field and rack

focus to videos. It is automatically able to shift focus between two subjects in a frame and also adds a subtle bokeh in the background. This mode records in Dolby Vision HDR and lets you edit the focus post-recording.

Talking about videos, iPhone cameras have always been great and the iPhone 13 is no different. Recording 4K HDR videos in Dolby Vision, makes videos from the iPhone 13 look very good and the colours are closer to natural tones. The sensor-shift OIS works well in reducing jittery videos.

At night, however, the iPhone cameras have to catch up with Android phones as the pictures often lack sharpness. Sadly the night mode doesn't live up to its potential.

The selfie camera gets the same features as the rear cameras and remains one of the best with spot on skin tones and details.



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BATTERY LIFE

The battery life of the iPhone 13 has improved as well and depending on your use you can get over 7 plus hours of screen-on time, which is impressive. And the iPhone 13 tops up in around 1 hr 40 mins using the standard

Type-C to Lightning adapter.

VERDICT

The iPhone 13 is an upgrade for sure. It uses new camera sensors, brighter display, faster A15 Bionic chip and offers better battery life. At its asking price of ₹79,900, the iPhone 13 is a no-brainer for someone who is looking to upgrade from the iPhone 11 or before.

—Siddharth Chauhan

SPECIFICATIONS

DISPLAY SIZE: 6.10-inch | DISPLAY TYPE: Super Retina XDR OLED | Display resolution: 1170 X 2532 PIXELS | PROCESSOR: Apple A15 Bionic | FRONT CAMERA: 12MP | REAR CAMERA: 12MP + 12MP | STORAGE: 128GB | OS: iOS 15 | WEIGHT: 174g | IP RATING: IP68 | BATTERY CAPACITY: 3,227mAh | FAST CHARGING: 23W

CONTACT

APPLE | PHONE: 0008001009009 | EMAIL: Web form | WEBSITE: <https://www.apple.com/in/>